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Wednesday LEOP Simulation — Instructor Pack

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Classification: Internal

Wednesday LEOP Simulation — Instructor Pack

Confidential — instructor and simulator operator only.

This pack covers the Wednesday simulation day. Two student teams swap operator/observer roles between two passes. Anomaly injection choices are listed; the instructor picks per team strength after Tuesday's training quiz.

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1. Day at a glance

When	What	Mode
09:00 – 09:30	Standup, scenario brief to both teams	Instructor-led
09:30 – 11:30	**Pass 1** — Team A operates; Team B observes	LEOP sim
11:30 – 12:30	Lunch + observer prep	—
12:30 – 12:45	Reset + brief for Pass 2	Instructor-led
12:45 – 14:45	**Pass 2** — Team B operates; Team A observes	LEOP sim
14:45 – 16:00	Joint debrief, observer presentations	Plenary
16:00 – 17:00	Anomaly-ticket completion + shift logs	Individual

Each pass is 90 min real time / 90 min sim time (1× speed).

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2. Scenario configuration

- **Breakpoint:** BP-2 — LEOP Day Start (existing scenario `scenarios/first_contact.yaml`).
- **Phase at start:** 3 (BOOTLOADER_OPS).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- `override_passes` flag set so contact remains for the pass duration.
- **Speed multiplier:** 1× (real time).

There is no `sim run` command. Start the simulator once (`smo-simulator --config configs/eosat1/ --speed 1`, or `./start.sh` for the whole suite), then start the scenario by the `name:` in its YAML — **First Contact** — from the instructor console's Scenarios panel, or over the instructor API on the simulator port:

```
curl -X POST http://localhost:8080/api/command \
     -H 'Content-Type: application/json' \
     -d '{"type": "start_scenario", "scenario": "First Contact"}'
```

Start an anomaly from §4 the same way, by its name, at its cue moment. Only one scenario is active at a time: starting the anomaly replaces *First Contact* as the active scenario but keeps the spacecraft state, and the anomaly's events run from that moment. `first_contact.yaml` declares `duration_s: 1800`; the pass runs for as long as the instructor keeps the clock going.

3. Procedure expectations

Operator team is expected to drive the spacecraft from phase 3 (bootloader, no antennas) to phase 4 (LEOP, full HK, detumble complete) in roughly that order:

1. **LEOP-001** First Acquisition of Signal & OBC Boot
2. **LEOP-002** Initial Health Check
3. **LEOP-004** Solar Array Verification (wing deployment)
4. **LEOP-007** Sequential Power-On (heaters if needed, the AOCS suite line, magnetometer, magnetorquers and GPS, then DETUMBLE; the wheels stay disabled until COM-004)
5. **LEOP-005** Sun Acquisition (observes the automatic handover to COARSE_SUN)
6. **LEOP-006** Time Synchronisation
7. (If time permits) **LEOP-003** Initial Orbit Determination, which ends LEOP by declaring phase 5

Pass 1 typical end-state: phase 4, antennas deployed, app SW running, htr_bat ON, aocs_wheels ON, body rates < 0.05 °/s.

Pass 2 typical end-state: same, in the COARSE_SUN sun hold, clock set; the payload stays off until COM-009; ready for commissioning Thursday.

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4. Anomaly injection menu

Pick **one** primary injection per team and one optional secondary ("rolling injection") if the primary is resolved cleanly. Files named are scenario YAML files in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`; trigger them through the scenario operator UI mid-run.

Severity	Scenario file	What students see	Recovery procedure
Easy	`ttc_no_tm_at_aos.yaml`	Beacon present, full HK never reports despite OBC_BOOT_APP	CTG-019 No TM at Pass Start
Easy	`eps_overcurrent.yaml`	Sudden overcurrent on a power line during sequential power-on	CTG-012 Overcurrent Response
Medium	`aocs_actuator_stuck.yaml`	One reaction wheel locked at 0 RPM during detumble	CTG-007 Reaction Wheel Anomaly
Medium	`aocs_star_tracker_failure.yaml`	Primary star tracker degrades at end of detumble	CTG-008 Star Tracker Failure
Hard	`obc_watchdog.yaml`	OBC watchdog reset mid-deploy	CTG-010 OBDH Watchdog Recovery
Hard	`obc_bus_failure.yaml`	CAN bus A drops mid-sequential-power-on	CTG-017 Bus Failure Switchover
	`gs_antenna_failure.yaml`		

Severity	Scenario file	What students see	Recovery procedure
Very hard		Iqaluit antenna degraded mid-pass; team must defer rest of pass to Troll	CTG-020 Ground Station Antenna Failure

All seven files exist in `configs/eosat1/scenarios/` (checked against the simulator tree on 2026-10-30). The console lists them by their `name:` field: *No Telemetry at AOS*, *LCL Overcurrent Trip*, *Reaction Wheel Stuck at Speed*, *Star Tracker Failure*, *OBC Watchdog Reset*, *CAN Bus A Failure* and *Ground Station Antenna Failure*. The instructor book's catalogue (§5.10.2) gives CTG-026 alongside CTG-007 for `aocs_actuator_stuck.yaml`.

Injection cue card

For each injection, the instructor needs to know:

1. **Cue moment** — at what point in the LEOP procedure to inject.
2. **First indicator** — what the operators will see first (telemetry parameter, FDIR alarm, voice cue).
3. **Pass criterion** — what "successful recovery" looks like for the pass.
4. **Five debrief questions** to use after the pass.

The cue cards for all seven are in the instructor book, chapter 18: §18.1 No TM at Pass Start, §18.2 Power-Line Overcurrent, §18.3 Reaction Wheel Stuck, §18.4 Star Tracker Failure, §18.5 OBC Watchdog Reset, §18.6 OBC Bus Failure and §18.7 Ground Station Antenna Failure. Each card gives the cue moment, first indicator, recovery procedure, pass criterion and five debrief questions. Print the cards you will use before the shift.

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5. Pass operator-side criteria

End-of-pass GO/NO-GO criteria (instructor uses for assessment):

Criterion	Pass 1 (Team A)	Pass 2 (Team B)
OBC application SW running	✓	✓
Onboard time set, drift < 0.5 s	✓	✓
Antennas deployed, high-rate link	✓	✓
Battery heater ON	✓	✓
Reaction wheels ON, AOCS in DETUMBLE	✓	✓
Body rates < 0.05 °/s	optional	✓
Anomaly ticket filed for any injection	✓	✓
FDIR rule sets enabled and verified	—	✓

Each criterion contributes to the team's **simulation performance** mark (Week 3 assessment row).

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6. Coaching during the shift

The instructor (acting as Mission Director) sits behind the FD seat and coaches by exception only:

- **Voice-loop discipline.** If the team starts cross-talking, stop, name the rule, restart.
- **Procedure compliance.** If a position commands without FD GO, stop, log it, treat as a safety violation.
- **Anomaly response.** If the team is stuck > 5 minutes, drop a hint at the FDIR-panel level (don't name the procedure).
- **Time pressure.** Maintain the real-time clock — no pausing the sim for discussion. Pass closes when the orbit closes.

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7. Debrief (joint)

Structure of the 75-minute debrief at end of day:

1. **Hot-wash (15 min).** Each team's FD reads their shift log highlights. No critique yet.
2. **Anomaly walk-through (20 min).** Operator team walks through the anomaly response; observer team challenges.
3. **Peer-review worksheet readout (20 min).** Each observer team reads its four-axis worksheet aloud (team / comms / space / technical). Operator team responds with one accept and one defend per axis.
4. **Lessons-learned synthesis (15 min).** Instructor consolidates into a single list. Logged in the cohort lessons-learned register.
5. **Brief Thursday (5 min).** Reading task: at minimum the COM-001 through COM-008 procedure cards before tomorrow.

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8. Safety and red lines

- **No commanding without FD authorisation.** Violation → instructor calls the shift, team logs the violation, no penalty if learnt-from.
- **No simulator-pause to win an exercise.** The clock runs.
- **No silent FD.** The Flight Director must be on the loop. If the FD is over-driven, the instructor calls a 30-second hold and resets the position.

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9. References & files

- Simulator scenario: `configs/eosat1/scenarios/first_contact.yaml`
- Anomaly scenarios: see §4 table
- Procedure files: `configs/eosat1/procedures/leop/*.md`
- MCS positions: `configs/eosat1/mcs/positions.yaml`
- Telemetry limits: `configs/eosat1/mcs/limits.yaml`
- ECSS-E-ST-70-11C — Space segment operability
- NASA *Mission Operations Handbook* — https://nodis3.gsfc.nasa.gov/
- ESA *Mission Operations Concept* — https://www.esa.int/Enabling_Support/Operations